

Christina Wang

Lederman Fellow at Fermilab

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Last updated: September 10,
2026

I search for new physics beyond the Standard Model along two fronts: pushing past detector limitations at the CMS experiment and developing novel detector technologies for future particle physics experiments. I pioneered a new experimental technique in CMS to detect long-lived particles invisible to conventional analysis, now used across 5+ CMS analyses. I have since improved its sensitivity with a new hardware-level trigger in Run 3 of the CMS experiment, delivering the first public results with the new trigger. In parallel, I helped build a new detector instrumentation lab at Fermilab from the ground up to characterize superconducting nanowire single-photon detectors, leading the first characterizations of these novel detectors for photons and high energy particles. Going forward, I plan to keep pushing these limits at the LHC as the dataset grows, while developing new detector technologies for the next generation of experiments.

RESEARCH INTERESTS

Experimental Particle Physics at the LHC; Search for long-lived particles and unconventional signatures; precision timing detectors; Quantum sensor for HEP; Superconducting nanowire single-photon detectors (SNSPDs)

EMPLOYMENT & EDUCATION

Lederman Fellow at Fermilab	September 2024 – present
Postdoctoral Scholar Research Associate at Caltech	April 2024 – September 2024
California Institute of Technology	October 2018 – March 2024
Ph.D. in Physics. Advisor: Maria Spiropulu	
Thesis: Search for New Physics with the CMS Experiment and QIS-enabled Technology	
Massachusetts Institute of Technology	September 2015 – June 2018
B.S. in Physics	

FELLOWSHIPS AND AWARDS

APS Mitsuyoshi Tanaka Dissertation Award in Experimental Particle Physics	2026
CMS Thesis Award	2024
DOE Graduate Student Research Fellowship (SCGSR)	2022
2022 Snowmass Poster Award	2022
Caltech Rosenbaum-Faber Family Fellowship	2018

APPOINTED POSITIONS

CMS Long-lived Particle Subgroup Convener	2024 – 2026
Reviewing and managing 20+ active long-lived particle analyses. 17 public results during tenure.	
CMS Exotica Methods Forum Contact	2023 – 2024
Started monthly meeting, documentation website, and common EXONanoAOD data format to foster exchange and preservation of unconventional techniques unique to exotica searches	

GRANTS AND FUNDING

Fermilab LDRD	2025
Superconducting Nanowire Detectors for High-Energy Particle Detection	
Fermilab Detector R&D New Initiatives	2024
Superconducting Nanowire Detectors for High-Energy Particle Detection	

RESEARCH EXPERIENCE

CMS Experiment

2017 – present

Searches for Long-lived Particles (LLPs) Decaying in the CMS Muon Detectors (2019–present)

Founder of the CMS muon detector shower (MDS) search program.

- Discovered the MDS signature and developed the reconstruction and identification techniques that established it as a powerful tool for LLP searches
- Led the first LLP searches using MDS, achieving the best sensitivity for LLPs with proper decay lengths from 5–10000 m using data collected from 2016–2018 [12], [20]
- Contributed to additional searches using the MDS signature for heavy neutral leptons, b-hadron decays, and vector-like leptons, achieving the best sensitivity for LLPs with proper decay lengths from 0.1–100 m [13], [6], [3]
- Designed and developed the MDS trigger for Run 3, improving trigger efficiency by an order of magnitude [1], [18], [19]
- Led an analysis team to analyze 2022–2024 data with the new MDS trigger [7]

Reinterpretation of CMS MDS Search to New Physics Models (2021–present)

- Developed a [new Delphes module for MDS](#) to enable future reinterpretations based on the parameterized detector response function measured for the first CMS endcap MDS search
- Recast the first CMS endcap MDS search, showing its complementarity to other LLP experiments for many LLP models [14], [17]

Search for Heavy Stable Charged Particles (HSCP) (2022–2023)

- Searched for HSCP with highly ionizing signature, setting stringent limits on the masses of gluinos up to 2 TeV and tau sleptons up to 0.69 TeV [9]
- Studied sensitivity of an additional double-HSCP analysis category and calculated systematic uncertainties of key variables and trigger efficiency scale factor

Search for LLPs Using Out-of-time Trackless Jets (2018–2023)

- Achieved the best sensitivity for long-lived neutralinos with proper decay lengths of 0.3–1.0 m, excluding masses up to 1.18 TeV [15]
- Developed the key discriminating variables used as inputs to the DNN for LLP-jet tagging

Precision Timing: MIP Timing Detector (MTD) for HL-LHC Upgrade (2018–2024)

- Measured light yield of LYSO with different LYSO+SiPM geometries, set up the data acquisition system, and prepared hardware for test beams at FNAL [21]
- Performed test beam measurements to test the TOFHIR ASIC readout chip and radiation hardness of LYSO crystals at FNAL [21] and LANL [11]

Measurements of Differential Z Boson Production Cross Section (MIT Undergraduate Research, 2017–2018)

- Measured Drell-Yan differential cross section with focus on the low- p_T region [22]
- Optimized differential bins, performed statistical analysis and unfolding procedure, and presented at the CMS SMP (Standard Model Physics) group

Dark Matter and Quantum Sensing Experiment

2021 – present

R&D for Superconducting Nanowire Single Photon Detectors (SNSPDs) (2022–present)

Built the Fermilab SNSPD detector lab from the ground up, including multiple integration and characterization systems for large-area SNSPDs and full cryogenic and room-temperature infrastructure.

- First temperature-dependent and multi-channel measurements for large-area (mm^2) SNSPDs [5]
- First ever characterization of the SNSPD response to high-energy particles at the Fermilab Testbeam Facility [10] and at the CERN SPS Testbeam facility [4]

BREAD: Broadband Reflector Experiment for Axion Detection (2022–2024)

- Search for axions using a parabolic reflector to focus axion-converted photons onto SNSPDs, achieving the best sensitivity in the meV–eV mass range

Dark Matter Detection with GaAs + SNSPDs (2022–2024)

- Prototype to search for sub-GeV dark matter candidates by coupling SNSPDs with gallium arsenide (GaAs), a bright cryogenic scintillator
- Developed an x-ray calibration system for the detector prototype with a Co57 source and high-purity germanium detectors to detect Compton-scattered photons
- Benchmarked performance (time jitter, synchronization, and noise metrics) of various commercial digitizers for the data acquisition system to inform procurement decisions

PUBLICATIONS & PUBLIC RESULTS

As a member of the CMS Collaboration, I am an author on 500+ publications [INSPIRE-HEP]. Below is a selected list of publications for which I made significant contributions.

1. M. Abbott et al. “High Multiplicity Trigger for Long-Lived Particles in CMS detector.” [JINST 21 P07040 \(2026\)](#)
2. CMS Collaboration. “Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at $\sqrt{s} = 13.6$ TeV.” [Submitted to Phys. Rept.](#)
3. CMS Collaboration. “Search for b-hadron decays to long-lived particles in the CMS endcap muon detectors.” [Phys. Rev. D 113, 012009 \(2026\)](#)
4. C. Wang et al. “Towards high-efficiency particle detection using superconducting microwire arrays.” [JINST 20 P12033](#)
5. C. Wang et al. “Temperature-Dependent Characterization of Large-Area Superconducting Microwire Array with Single-Photon Sensitivity in the Near-Infrared.” [JINST 20 P08005](#)
6. CMS Collaboration. “Search for vector-like leptons with long-lived particle decays in the CMS muon system in proton-proton collisions at $\sqrt{13}$ TeV.” [JHEP 08 \(2025\) 156](#)
7. CMS Collaboration. “Search for long-lived particles with a pair of muon detector showers in proton-proton collisions at $\sqrt{s} = 13.6$ TeV.” CMS Physics Analysis Summary [CMS-PAS-EXO-25-005](#), 2025.
8. CMS Collaboration. “Dark sector searches with the CMS experiment.” [Phys. Rept. 1115, 448 \(2025\)](#)
9. CMS Collaboration. “Search for heavy long-lived charged particles with large ionization energy loss in proton-proton collisions at $\sqrt{s} = 13$ TeV.” [JHEP 04 \(2025\) 109](#)
10. C. Peña et al. “High Energy Particle Detection with Large Area Superconducting Microwire Array.” [JINST 20 P03001](#)
11. C. Wang et al. “Radiation damage in 20-cm long LYSO:Ce and BaF₂:Y crystals.” [IEEE Transactions on Nuclear Science, vol. 71, no. 9, 2024](#)
12. CMS Collaboration. “Search for long-lived particles decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{13}$ TeV.” [Phys. Rev. D 110, 032007 \(2024\)](#)
13. CMS Collaboration. “Search for long-lived heavy neutral leptons decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s} = 13$ TeV.” [Phys. Rev. D 110, 012004 \(2024\)](#)
14. A. Mitridate et al. “Energetic long-lived particles in the CMS muon chambers.” [Phys. Rev. D 108, 055040 \(2023\)](#)
15. CMS Collaboration. “Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at $\sqrt{13}$ TeV.” [JHEP 07 \(2023\) 210](#)

16. S. Xie et al. “Entangled Photon Pair Source Demonstrator using the Quantum Instrumentation Control Kit System.” [IEEE Journal of Quantum Electronics](#), vol. 59, no. 5, 2023
17. G. Cottin et al. “Long-lived heavy neutral leptons with a displaced shower signature at CMS.” [JHEP 02 \(2023\) 011](#)
18. CMS Collaboration. “Performance of long lived particle triggers in Run 3.” CMS Detector Performance Summaries [CMS-DP-2023-043](#), 2023.
19. CMS Collaboration. “CSC High Multiplicity Trigger in Run 3.” CMS Detector Performance Summaries [CMS-DP-2022-062](#), 2022.
20. CMS Collaboration. “Search for long-lived particles decaying in the CMS endcap muon detectors in proton-proton collisions at $\sqrt{13}$ TeV.” [Phys. Rev. Lett. 127, 261804 \(2021\)](#)
21. The CMS MTD Collaboration. “Test beam characterization of sensor prototypes for the CMS Barrel MIP Timing Detector.” [JINST 16 \(2021\) P07023](#)
22. CMS Collaboration. “Measurements of differential Z boson production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV.” [JHEP 12 \(2019\) 061](#)

SEMINARS

Search for New Physics with CMS and Quantum Sensors

LBNL

October 2023

Search for Neutral Long-lived Particle Decaying in the CSCs

UNM NUPAC Seminar (Virtual)

October 2021

TALKS & POSTERS

CMS Experiment

Searches for Long-lived Particles and Unconventional Signatures with CMS

International Conference on High Energy Physics (Natal, Brazil)

August 2026

Search for Vector-like Leptons and Vector-like Quarks in CMS

Moriond Electroweak Interactions & Unified Theories (La Thuile, Italy)

March 2025

Long-lived Particles in the Muon Detectors

14th LHC LLP Workshop (Tokyo, Japan)

July 2024

Proton-Induced Radiation Damage in 20 cm Long LYSO:Ce and BaF₂:Y Crystals (Poster)

2023 IEEE NSS MIC RTSD (Vancouver, Canada)

November 2023

Search for New Physics with Long-lived and Unconventional Signatures in CMS

SUSY2023 (Southampton, UK)

July 2023

Invisible Higgs Decay at CMS

Standard Model at the LHC Workshop (Batavia, IL)

July 2023

Reinterpretation for Neutral Long-lived Particle Decaying in the CSCs

LHC Reinterpretation Forum (CERN)

December 2022

12th LHC LLP Workshop (Virtual)

November 2022

APS April Meeting (New York, NY)

April 2022

10th LHC LLP Workshop (Virtual)

November 2021

RAMP Seminar Series (Virtual)

September 2021

Search for Neutral Long-lived Particle Decaying in the CSCs

Poster at Snowmass Summer Workshop (Seattle, WA)

July 2022

9th LHC LLP Workshop (Virtual)

May 2021

APS April Meeting (Virtual)

April 2021

Status and Prospect of Search for LLP at ATLAS and CMS

Mitchell Conference (College Station, TX)

May 2022

Dark Matter and Quantum Sensing Experiment

High Energy Particle Detection with Large Area Superconducting Microwire Array

Applied Superconductivity Conference 2026 (Pittsburgh, PA) (Invited)	September 2026
APS Joint March Meeting and April Meeting (Denver, CO)	March 2026
CPAD (Philadelphia, PA)	November 2025
Annual Meeting of the International Muon Collider Collaboration (Hamburg, Germany)	May 2025
APS Joint March Meeting and April Meeting (Anaheim, CA)	March 2025
CPAD (Nashville, TN)	November 2024

Searching for Axions and Dark Photons with SNSPDs in the BREAD experiment

CPAD (Nashville, TN)	November 2024
CPAD (Menlo Park, CA)	November 2023
APS April Meeting (Minneapolis, MN)	April 2023
CPAD (Stony Brook, NY)	November 2022

BREAD: Broadband Reflector Experiment for Axion Detection

Applied Superconductivity Conference 2024 (Salt Lake City, UT) (Invited)	September 2024
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FPGA Based Signal Generation and Readout of SNSPD Signals for Quantum Communication

APS March Meeting (Las Vegas, NV)	March 2023
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SERVICE

Meeting of the Division of Particles and Fields of APS 2026

- Instrumentation track convener and session chair

CMS Exotica Workshop 2024 – 2025

- Long-lived session co-organizer and co-chair

Co-organizer of Fermilab Research Techniques Seminar 2026 –

Journal referee for papers in Physical Review Letters 2025 –

TEACHING

Saturday Morning Physics 2025 – present

- Co-organizer for a biannual 11-lecture series aimed at high school students
- Lecturer for “Particle Physics at the LHC”

CMS Data Analysis School 2024 – 2026

- Created and led the first two-day [data analysis exercise](#) focusing on long-lived particles
- Introduce groups of 10 graduate students to data analysis with LHC data
- Student groups from the exercise won best presentation prizes for the past three years

OUTREACH

CU*iP at Fermilab 2024 – 2025

- APS Conferences for Undergraduate Women and Gender Minorities in Physics
- Presenter, session chair, and member of the local organizing committee

Fermilab YouTube Shorts and Instagram Reels for World Quantum Day 2023

Caltech FUTURE of Physics Program 2018 – 2022

- An annual two-day program that brings undergraduate women in physics to Caltech for mini-workshops on navigating graduate applications and graduate student life, particularly as a woman

- Served as a panelist, gave lab tours and presentations, and starred in a [five-minute video](#) in the pandemic edition of the program introducing my career path and student life at Caltech

MIT Educational Councilor (Greater LA Region)

2018 – 2022

- Conducted in-depth interviews of about 10 first-year undergraduate applicants per year and submitted reports to the MIT Admissions Office

MIT Code for Good (Co-Director)

2015 – 2018

- Organized an annual one-month January-term class, Code for Good, matching teams of 3–4 students with non-profit organizations for programming projects
- Organized lectures, office hours, and final project expositions, and secured sponsorships from Facebook, Microsoft, and Yahoo to fund the program

MIT Physpop Program

2016

- Organized a week-long Freshman orientation program in Physics for 50 incoming students
- Coordinated introductory lectures from MIT physics professors and lab tours

SUPERVISION AND MENTORSHIP

Ph.D. Students

- **Matias Barria** (Purdue). Higgs to Di-muon with CMS Run 3 data (2025–present).
- **Elise Sledge** (Caltech). Higgs to Di-muon with CMS Run 3 data (2025–present).
- **Ludovico Mori** (Caltech). High energy particle detection with SNSPD (2024–present).
- **Thomas Sievert** (Caltech). High energy particle detection with SNSPD (2024–present).
- **Alexander Albert** (Caltech). Search for LLPs with MDS using Run 3 data (2024–present).
- **Chang Sun** (Caltech). Search for LLPs with MDS using Run 3 data (2024–present).
- **Paul Simmerling** (Caltech). Search for LLPs with MDS using 2022 data (2022–2023).
- **Anthony Aportela** (UC San Diego). Search for LLPs with MDS using B-Parking dataset (2022–2023).
- **Aram Hayrapetyan** (Yerevan University). Search for LLPs with MDS using B-Parking dataset (2022–2024).

Masters Students

- **Valentina Vega** (Universidad Técnica Federico Santa María). Building cosmic muon veto for SNSPDs (2024–2025).
- **Claudio San Martín** (Universidad Técnica Federico Santa María). High energy particle detection with SNSPD (2024).

Undergraduates

- **Joseph Sorel** (University of Chicago). Optimizing time resolution of large area SNSPDs (2026 Summer).
- **Jiahui Xie** (Caltech). Studied the impact of different muon reconstruction algorithms for Higgs to di-muon. (2025 Summer).
- **Sherry Wang** (Caltech WAVE Fellow). Reconstruction of boosted di-jet from Higgs boson decay into four bottom quarks (2023 Summer).
- **Ariela Yashinsky** (URA-Fermilab: Undergraduate Women in STEM). Reconstruction of boosted di-jet from Higgs boson decay into four bottom quarks (2023 Summer).
- **Samantha Contreras** (Caltech WAVE Fellow, went on to UCLA PhD). Phenomenology study of MDS from LLPs in the dark shower model; Goldwater Scholarship and Caltech Vodopia-Hasson Poster Competition winner (2022 Summer).
- **Gabby Dituri** (Caltech, went on to medical school at Tulane University). Reconstruction of low-mass LLPs in the CMS muon system; poster at the National Collegiate Research Conference at Harvard (2021) (2019–2022).

- **Nathan Suri** (Caltech, went on to Yale PhD). Phenomenology study to search for quirks with the CMS tracker and MIP Timing Detector at HL-LHC; talk at the [11th LHC LLP Workshop](#) (2018–2022).
- **Dawn Tang** (Caltech, went on to Berkeley PhD). CMS MTD upgrade: measured and studied the light collection efficiency of scintillation crystals with different geometries and wrapping materials (2018–2019).
- **Linus Kim** (Caltech, went on to Berkeley PhD). CMS MTD upgrade: automated power supply settings and the data acquisition and logging procedure for test beams at FNAL (2018–2019).